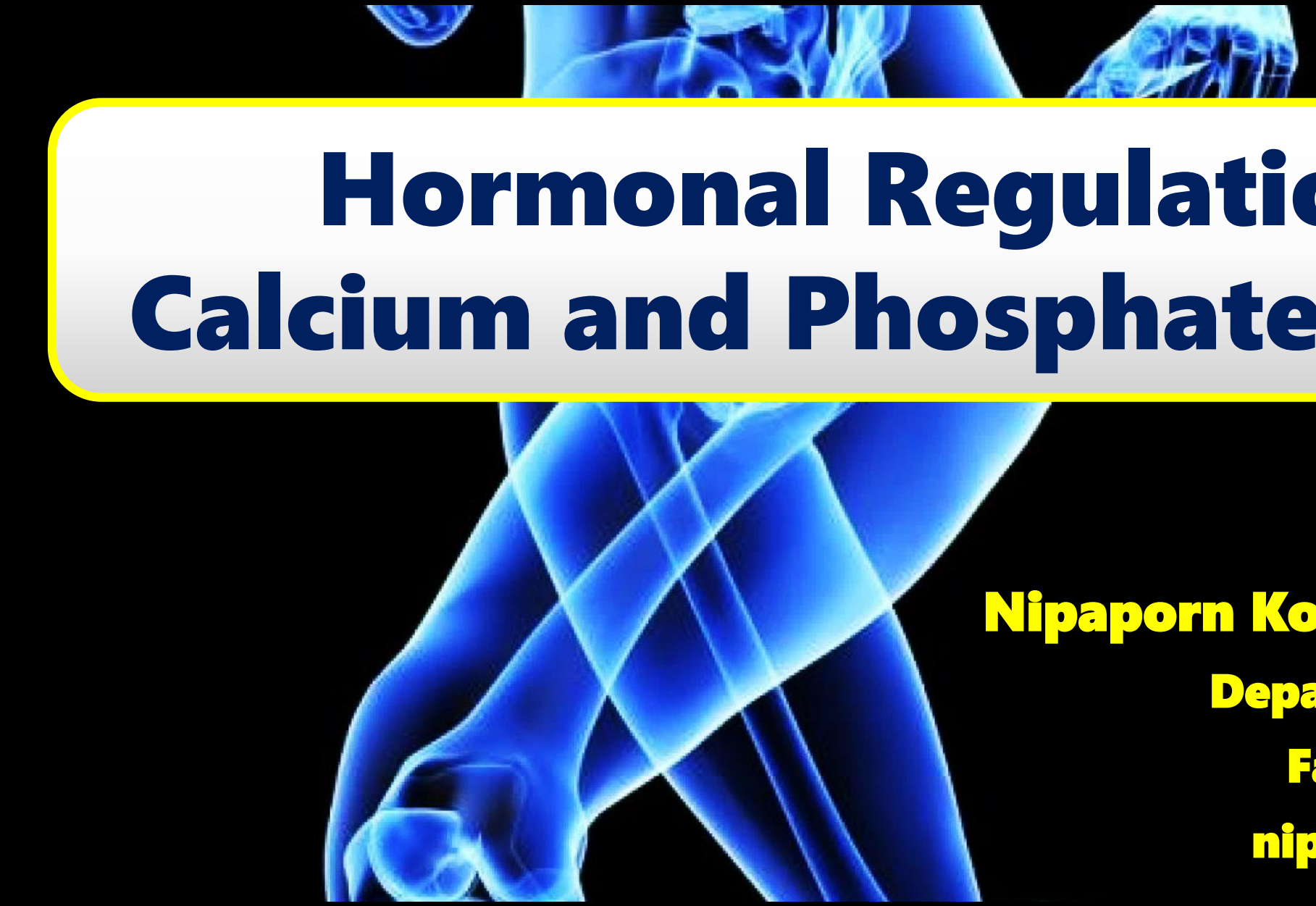


Hormonal Regulation of Calcium and Phosphate Balance

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Hormonal Regulation of Calcium and Phosphate Balance

วัตถุประสงค์ เมื่อสิ้นสุดการเรียนการสอน นักศึกษาสามารถอธิบาย

1. บทบาทของแคลเซียมและฟอสเฟตในร่างกาย
2. กลไกการควบคุมสมดุลของแคลเซียมและฟอสเฟต
3. บทบาทของฮอร์โมนที่ควบคุมสมดุลแคลเซียมและฟอสเฟตในร่างกาย

Content

1. Role of calcium and phosphate in the body
2. Calcium and phosphate homeostasis
3. Roles of hormones in regulating calcium and phosphate homeostasis
- **3.1 Vitamin D**
- **3.2 Parathyroid hormone**
- **3.3 Calcitonin**

Calcium (Ca^{2+})

- Ca^{2+} is an essential divalent cation.
- Obtain from the diet
- A component of the skeleton and teeth (>99%), complexed with P_i as **hydroxyapatite crystal [$Ca_{10}(PO_4)_6(OH)_2$]**



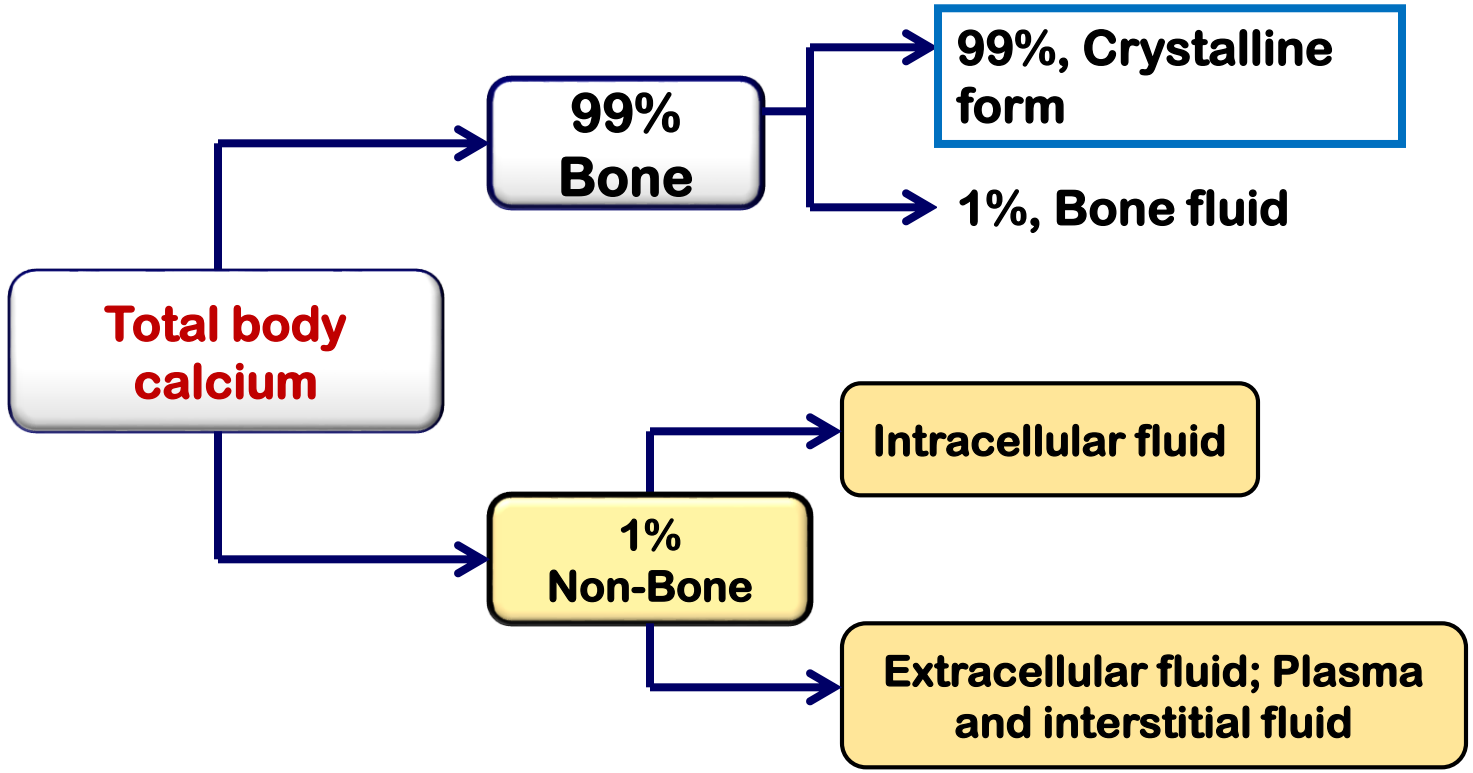
TABLE 13.1 Approximate Distribution of Calcium and Phosphorus in Adult Humans

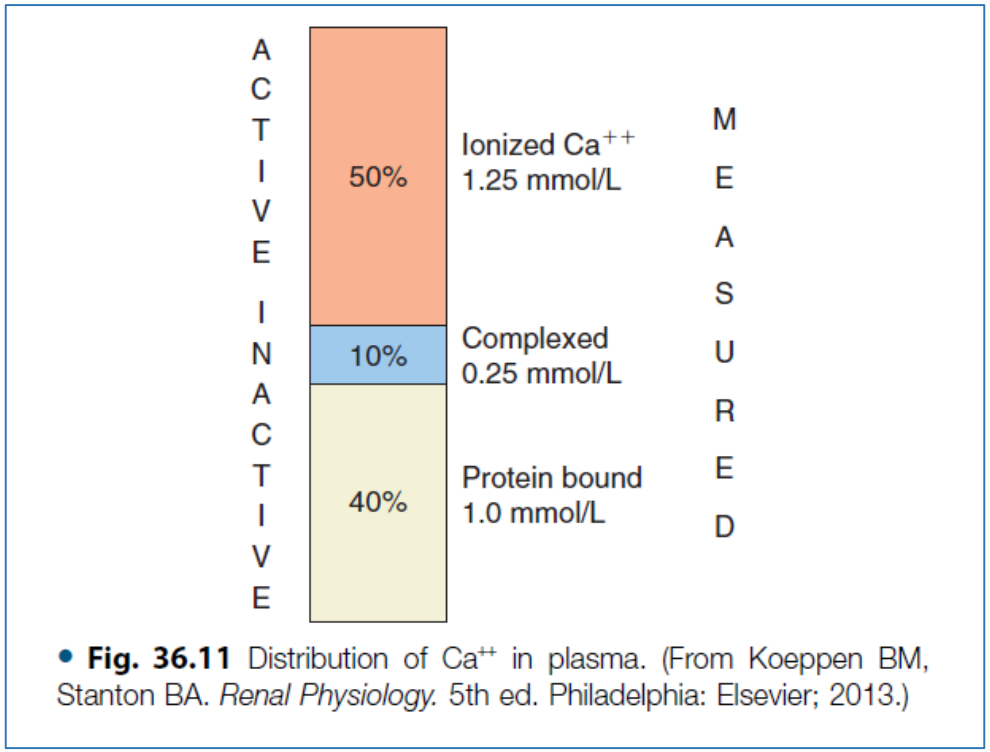
	Total Body Content (g)	% in Skeleton	% in Soft Tissue	% in Plasma
Calcium	1000	99	<1	<1
Phosphorus	700	85	14–15	<1

(DiMeglio and Imel, 2014)

Calcium distribution in the body

SDL





• **Fig. 36.11** Distribution of Ca^{++} in plasma. (From Koeppen BM, Stanton BA. *Renal Physiology.* 5th ed. Philadelphia: Elsevier; 2013.)

(Koeppen and Stanton, 2018)

- **In the plasma**, Ca^{2+} is present in 3 forms;
  - a biologically active ionized form (free form)
  - bound to proteins (primarily albumin)
  - a complex form with anions (e.g., citrate, carbonate, oxalate and P_i)
- **Intracellular Ca^{2+}** is in free and bound forms.

Functions of Ca^{2+} in the body

SDL

- **Structural roles in hard tissues** (i.e., bone and teeth)
- **Intracellular Ca^{2+}** is an important 2nd or 3rd messenger and cofactor of many enzymes and proteins necessary for normal cell functions.
- **Extracellular Ca^{2+}** is required for regulation of neuronal excitability, muscle contraction and relaxation, hormonal secretion, blood coagulation, and extracellular messenger (binding to the Ca^{2+} sensing receptor (CaSR) to activate cellular responses).
- Significant decrease in plasma Ca^{2+} can rapidly lead to death.
- A chronic increase in plasma Ca^{2+} can lead to soft tissue calcification and formation of stones.

Functions of Phosphate (P_i) in the body

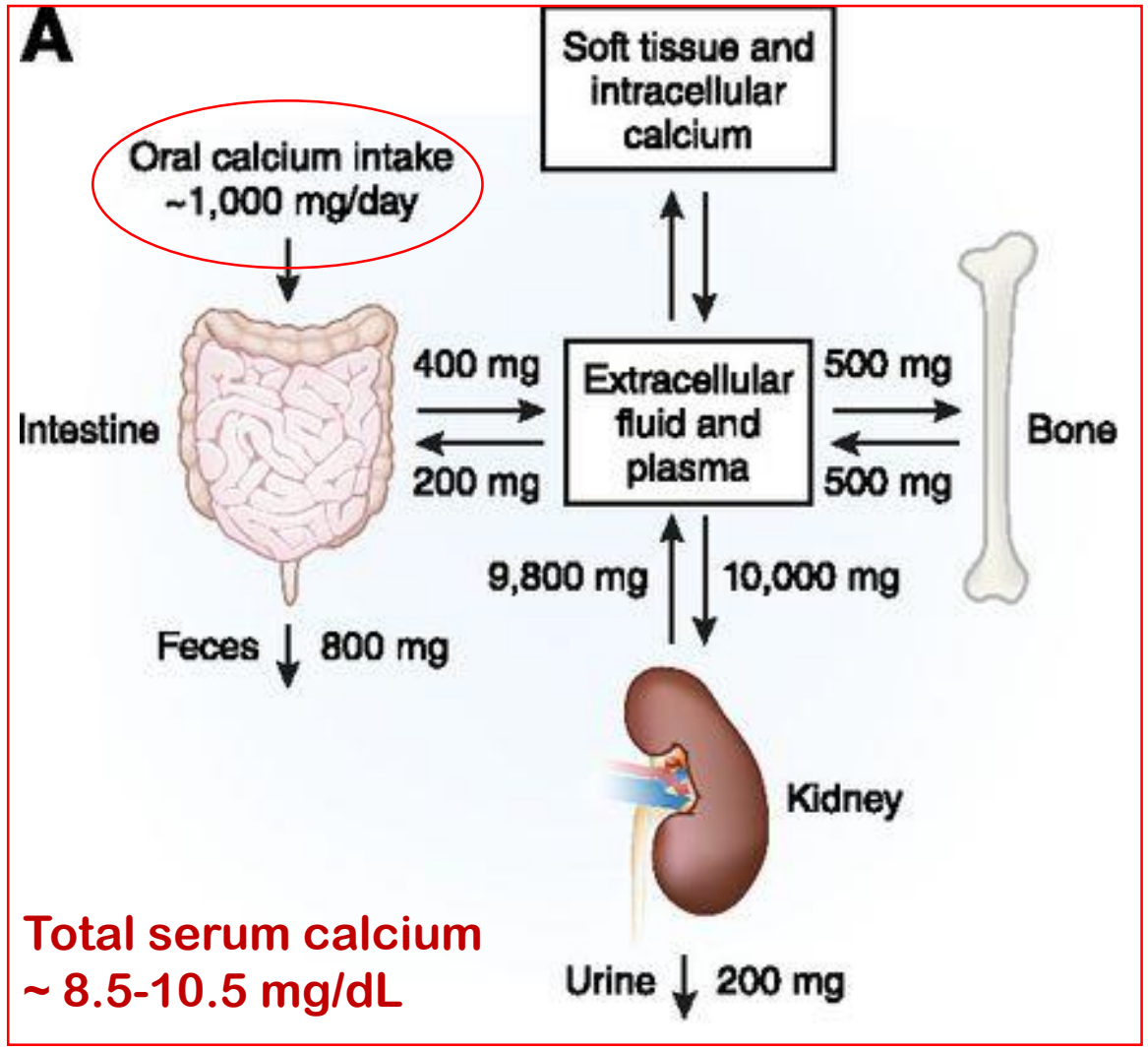
SDL

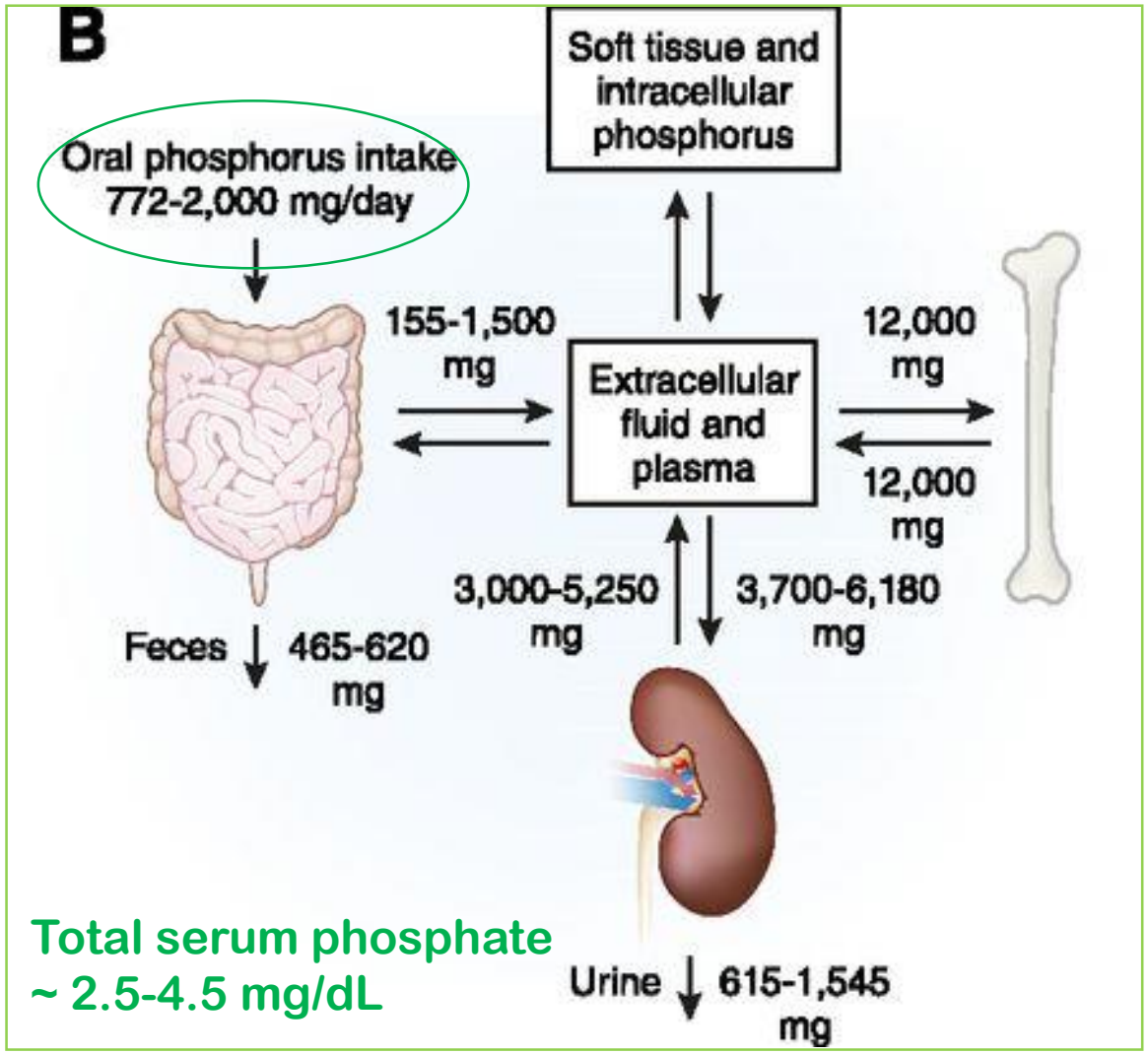
- **Phosphorus** is a key constituent of bone.
- Intracellular pH buffering and cellular energy metabolism as part of adenosine triphosphate (ATP)
- Macromolecules, such as nucleic acid, phospholipids, and phosphoproteins.

- Phosphorus circulates in the plasma primarily as inorganic orthophosphate (PO_4).
  At a normal blood pH of 7.4, 80% of the phosphate is in the HPO_4^{2-} form and 20% is in the $H_2PO_4^-$ form.
- In addition to free orthophosphate, P_i is present in small amounts in the plasma in organic form such as in hexose or in lipid phosphates.

Pi in plasma is ionized (45%),
complexed (30%) and bound to protein (25%).

Calcium and Phosphate Homeostasis

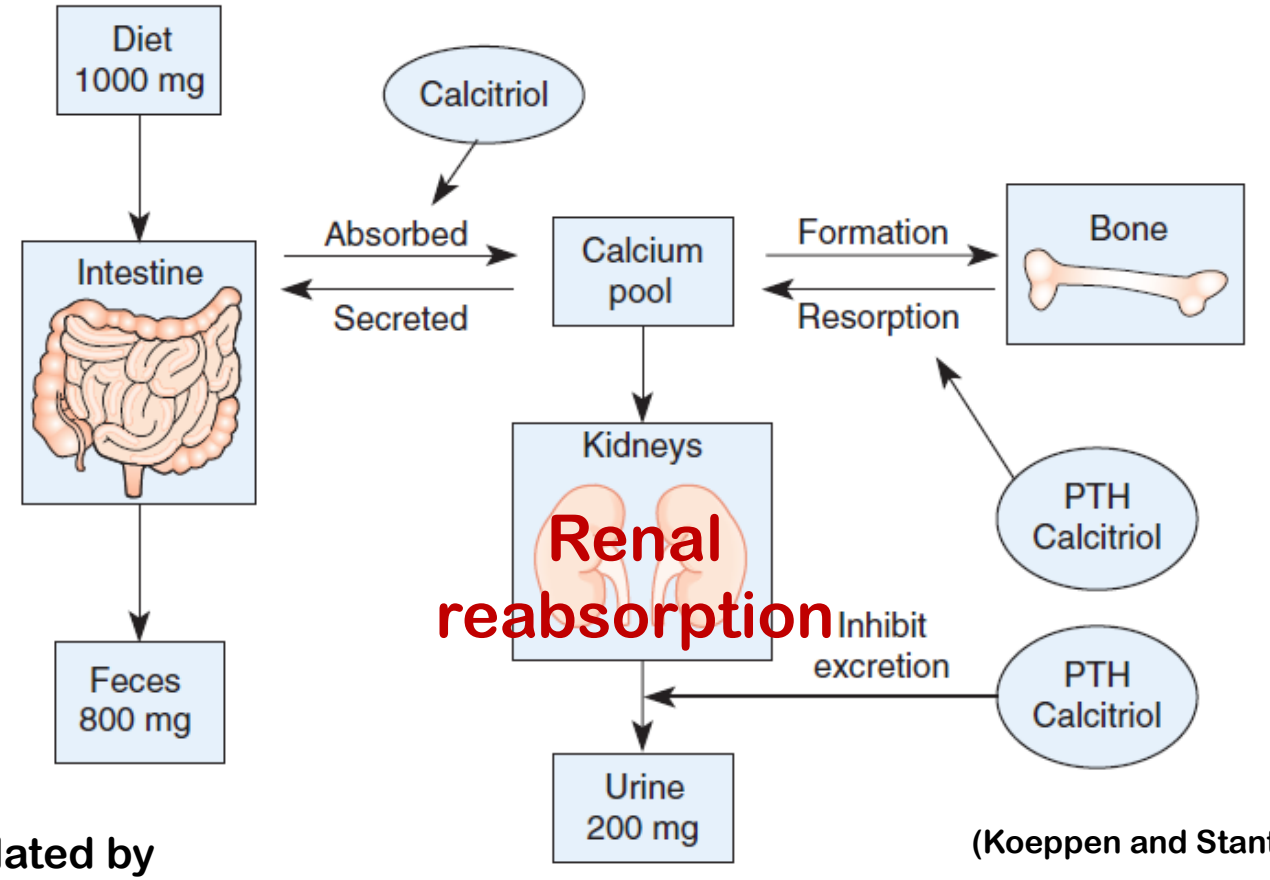




(Blaine et al., 2014)

Calcium Homeostasis

Intestinal absorption



Bone turnover

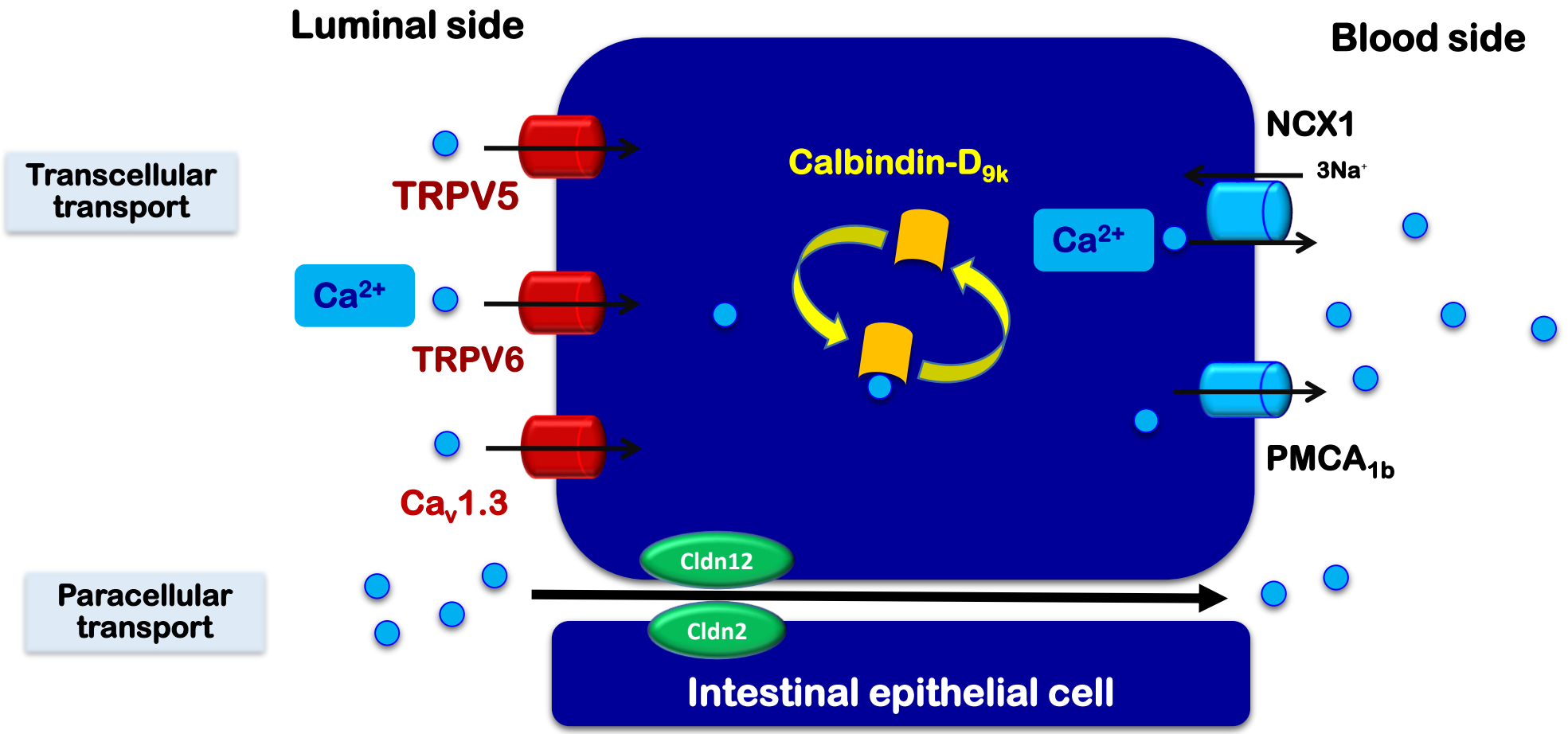
(Koeppen and Stanton, 2018)

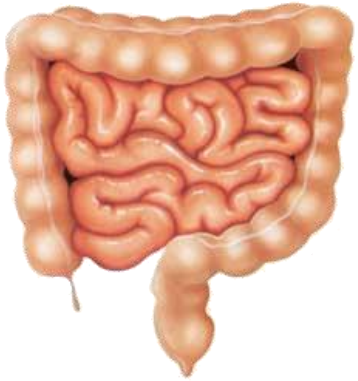
These mechanisms are regulated by the actions of the calciotropic hormones:

- Active form of vitamin D; 1,25-dihydroxyvitamin D [$1,25(OH)_2D_3$]
- Parathyroid hormone (PTH)
- Calcitonin

- Glucocorticoids, Estrogen, FGF-23 and plasma ionized Ca^{2+} concentration

Intestinal Calcium absorption

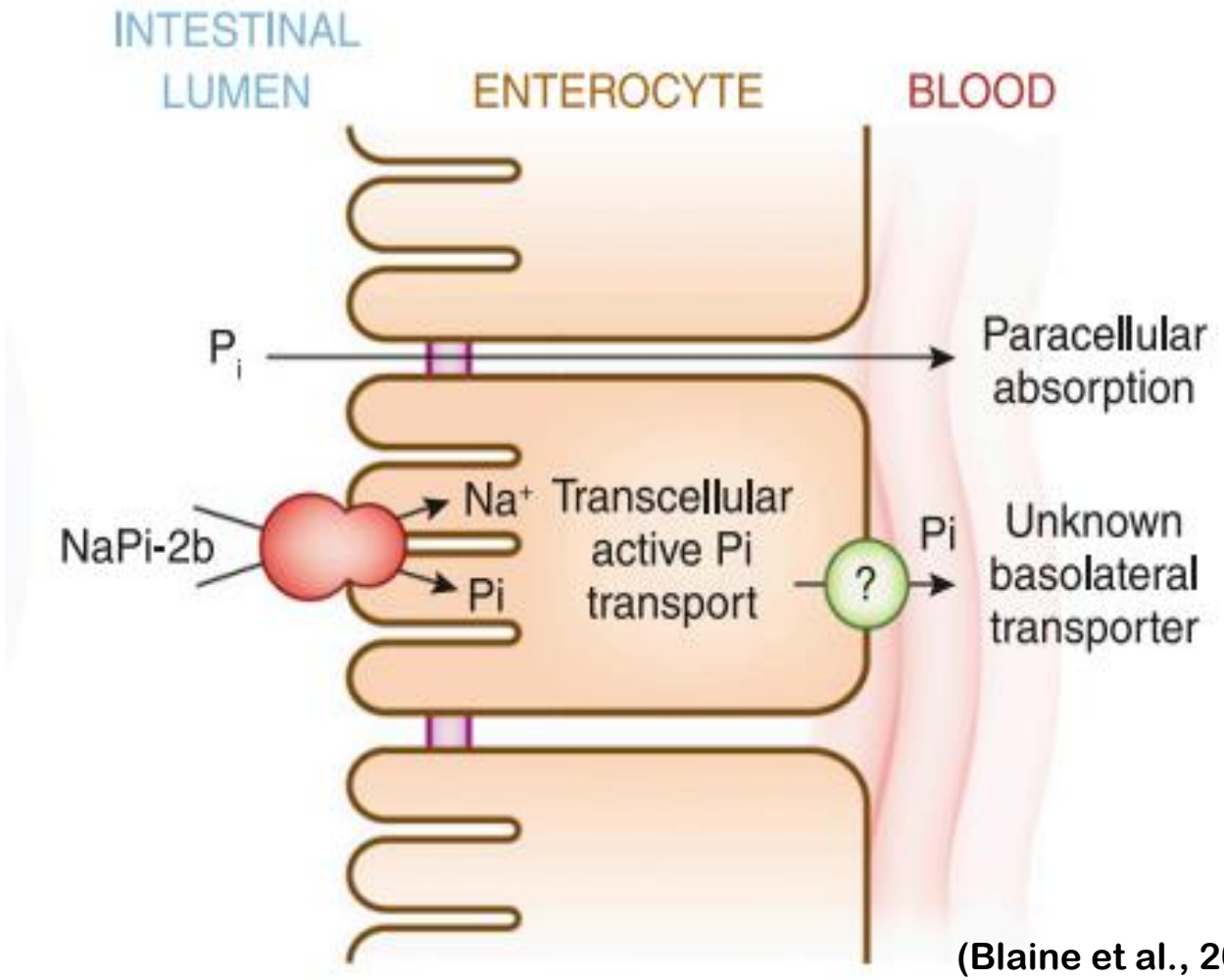


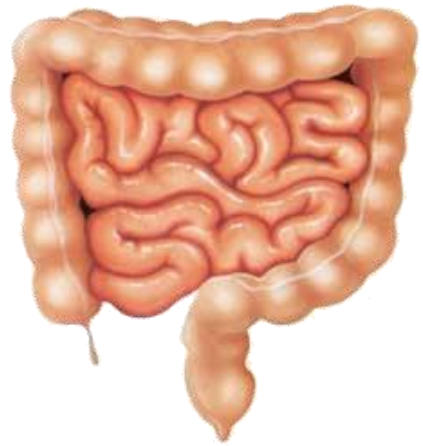


- Transient receptor potential vanilloid channels 5 and 6: TRPV5, TRPV6
- Calbindin-D_{9k}
- Plasma membrane Ca^{2+}-ATPase isoform 1b: $PMCA_{1b}$
- Na^{+}/Ca^{2+}-exchanger-1: NCX1
- Voltage-dependent calcium channel type 1.3: $Ca_v1.3$

- Cldn2 : Claudin2
- Cldn12 : Claudin12

Intestinal Phosphate absorption





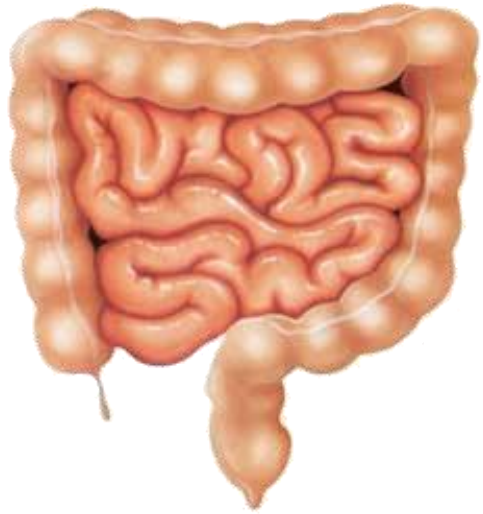
(Blaine et al., 2014)

- Sodium-phosphate cotransporter 2b (NPT2b) is the major transporter in the intestine.
- $1,25(OH)_2D_3$ increase expression of NPT2b.

Heterogeneity of the intestinal calcium transport

Adequate or high Ca^{2+} intake; (Paracellular Ca^{2+} transport)

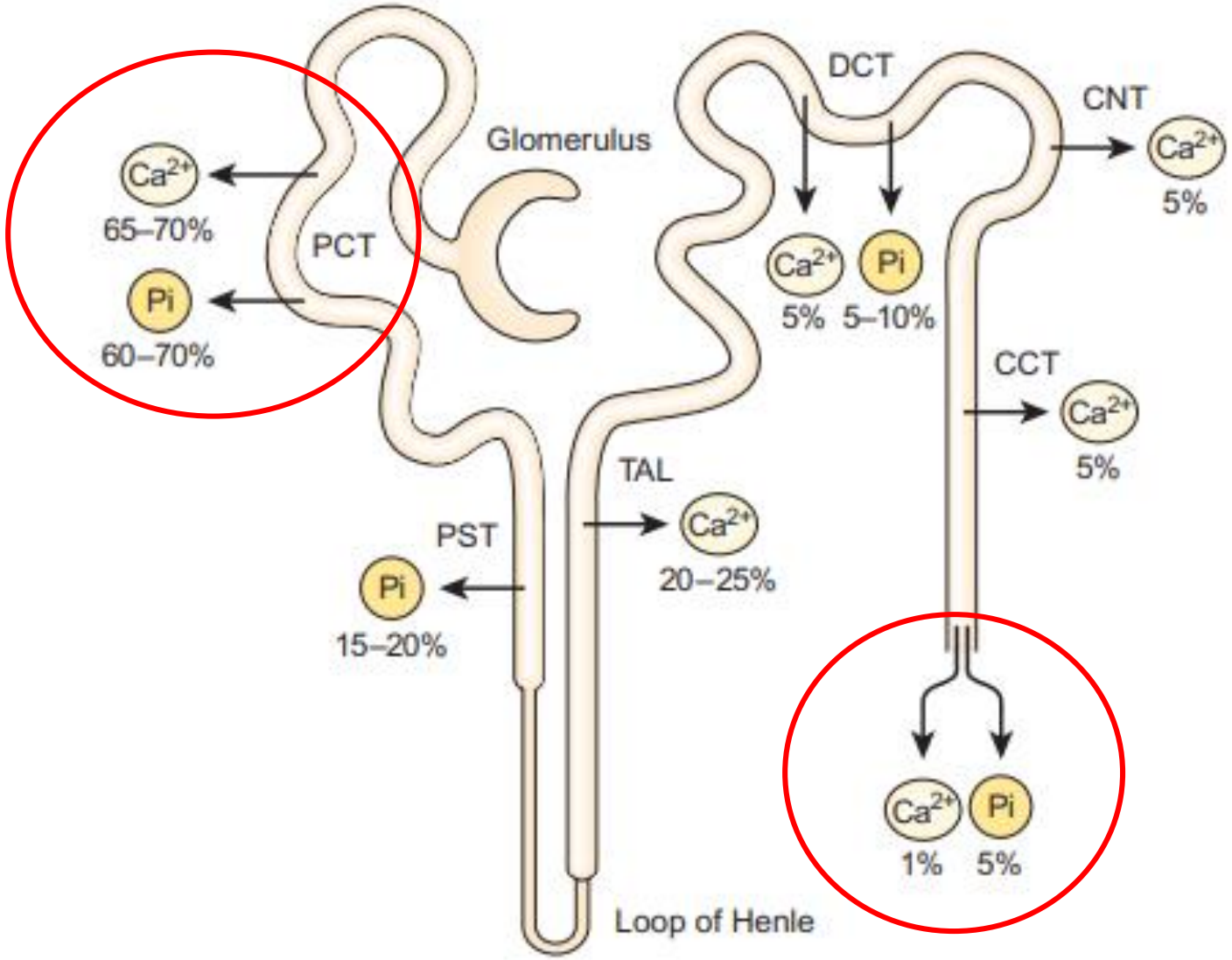
- 90% of the absorption the small intestine;
  - < 2% of Ca^{2+} absorption: Duodenum
  - ~25% of Ca^{2+} absorption: Jejunum
  - ~73% of Ca^{2+} absorption: Ileum (The longest transit time)
- 10% of the absorption the large intestine; cecum and colon



Low Ca^{2+} intake; (Transcellular Ca^{2+} transport)

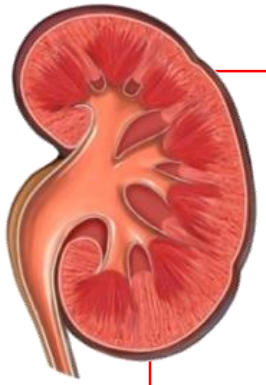
- Cecum is the site of the highest Ca^{2+} absorption rate per surface area.
  - highest mRNA expression of the key transcellular Ca^{2+} transporters (TRPV6 and calbindin-D_{9k})
- Duodenum is the most effective intestinal segment of Ca^{2+} absorption with the shortest transit time (~2 mins), followed by cecum and colon.

Renal reabsorption of Calcium and phosphate

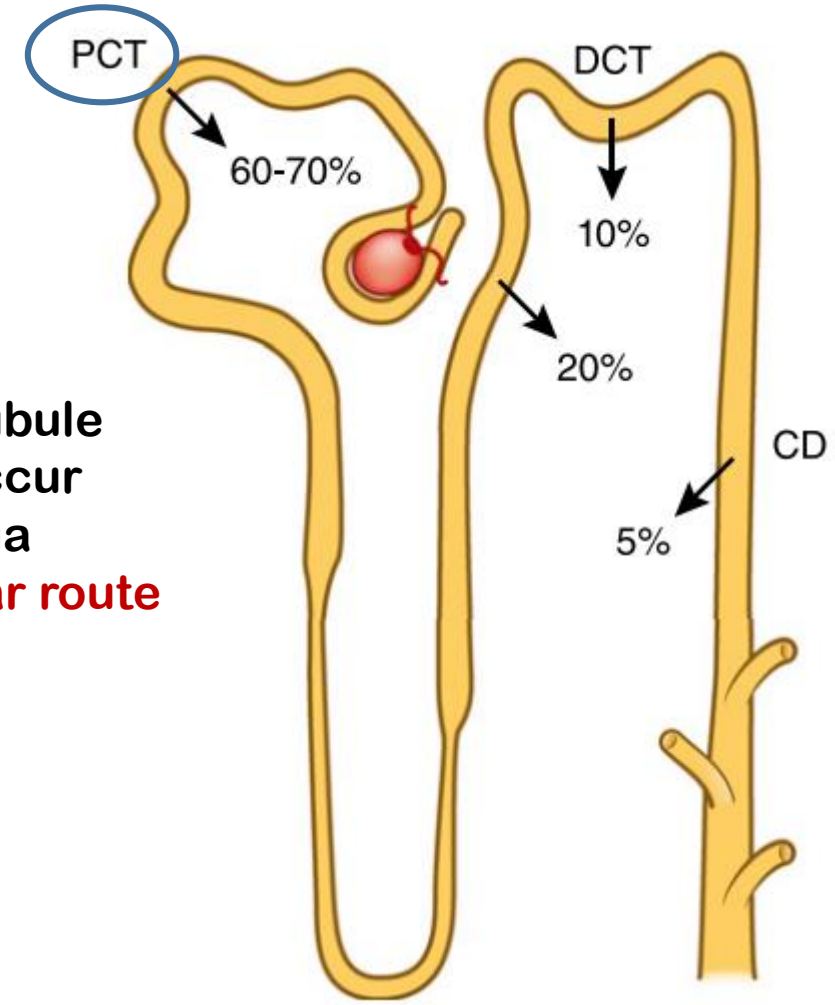


(DiMeglio and Imel 2014)

Renal reabsorption of Calcium and phosphate

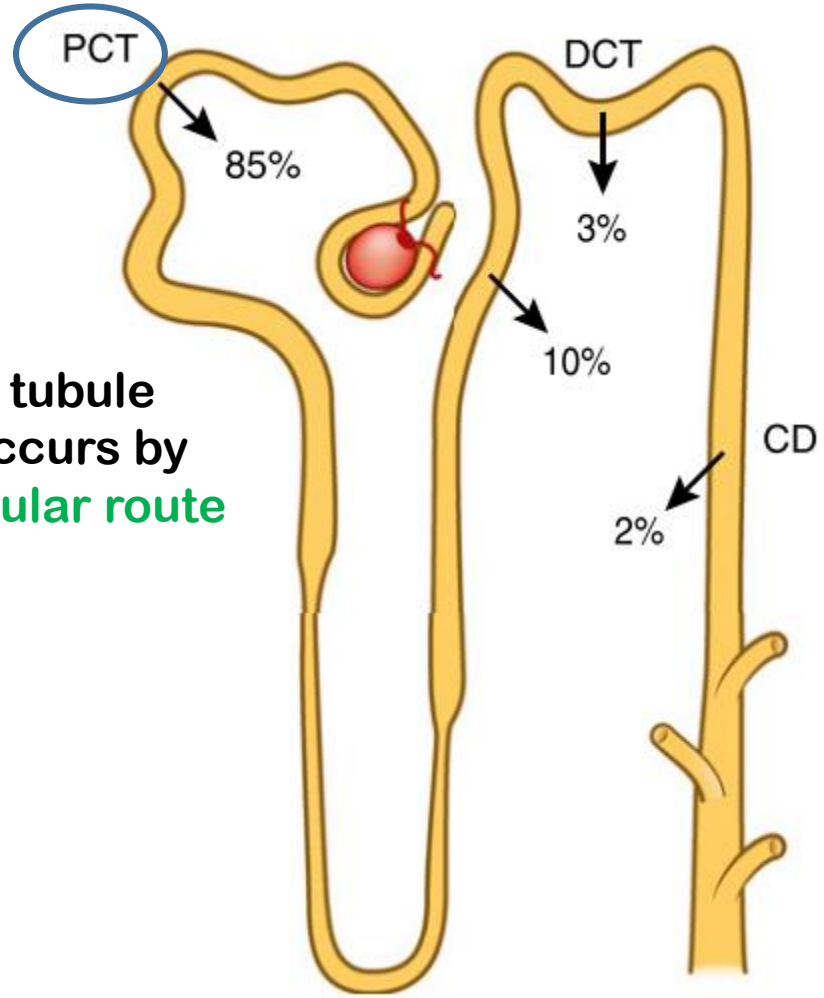


Ca^{2+}



Proximal tubule and TAL occur primarily via paracellular route

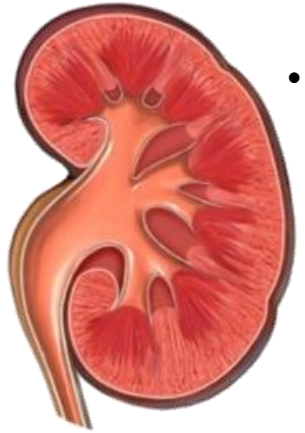
Pi



Proximal tubule mainly occurs by transcellular route

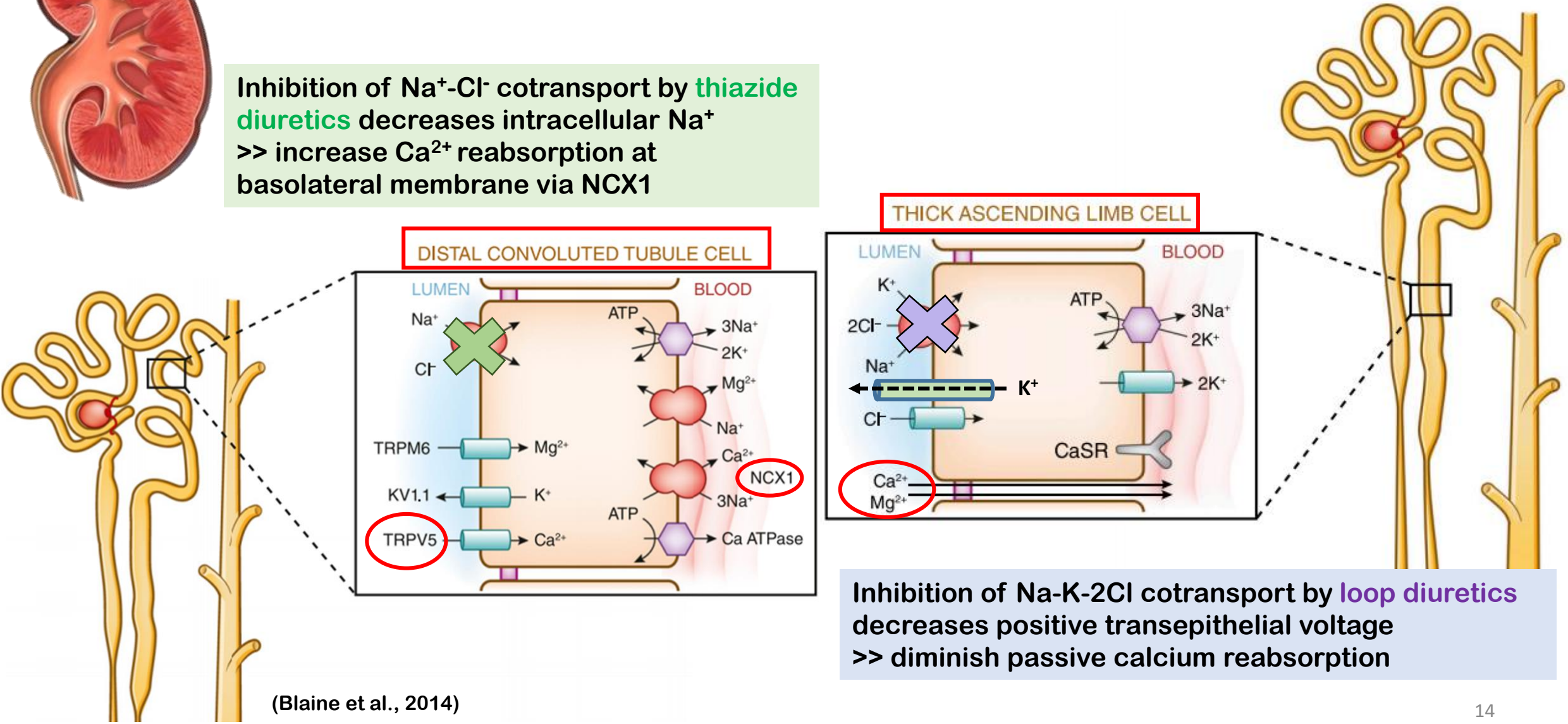
(Blaine et al., 2014)

Renal reabsorption of Calcium



- Basal absorption is passive transport and is driven by the electrochemical gradient for calcium.

Inhibition of Na^+-Cl^- cotransport by thiazide diuretics decreases intracellular Na^+ >> increase Ca^{2+} reabsorption at basolateral membrane via NCX1



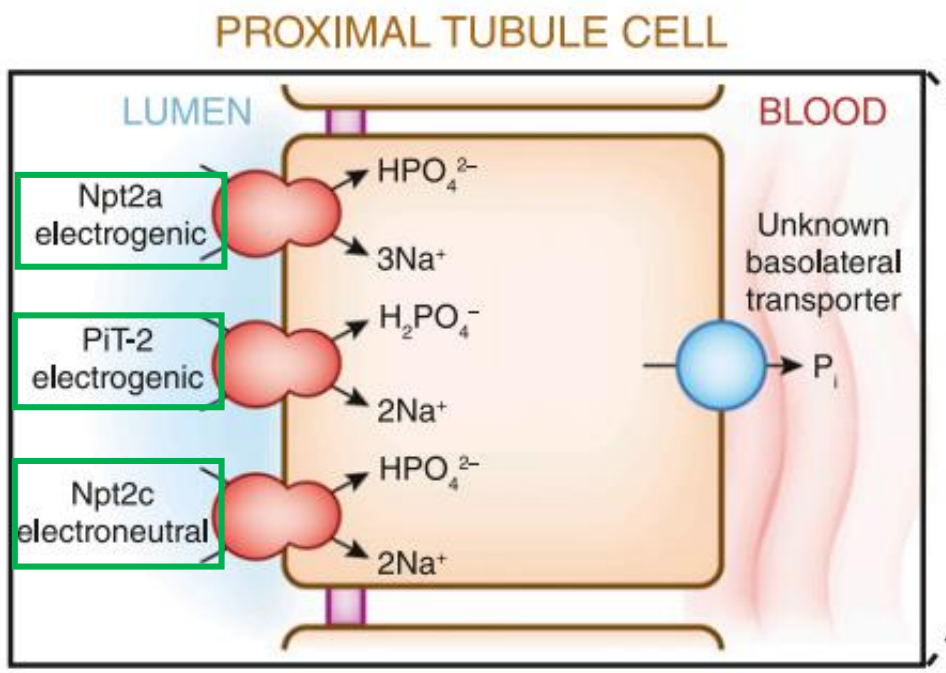
Inhibition of Na-K-2Cl cotransport by loop diuretics decreases positive transepithelial voltage >> diminish passive calcium reabsorption

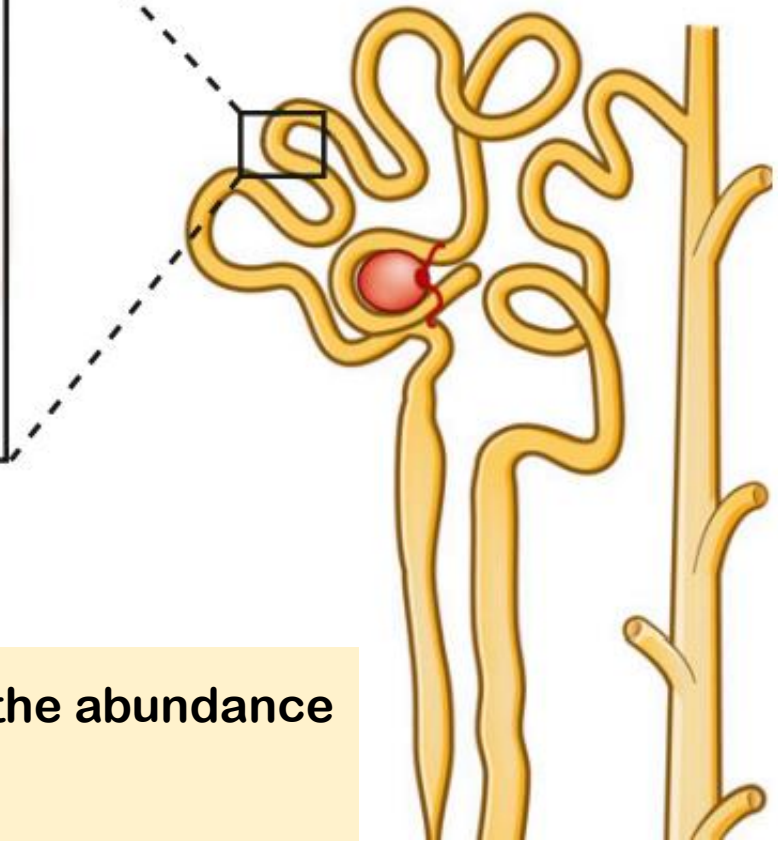
(Blaine et al., 2014)

Renal reabsorption of phosphate



Transcellular route

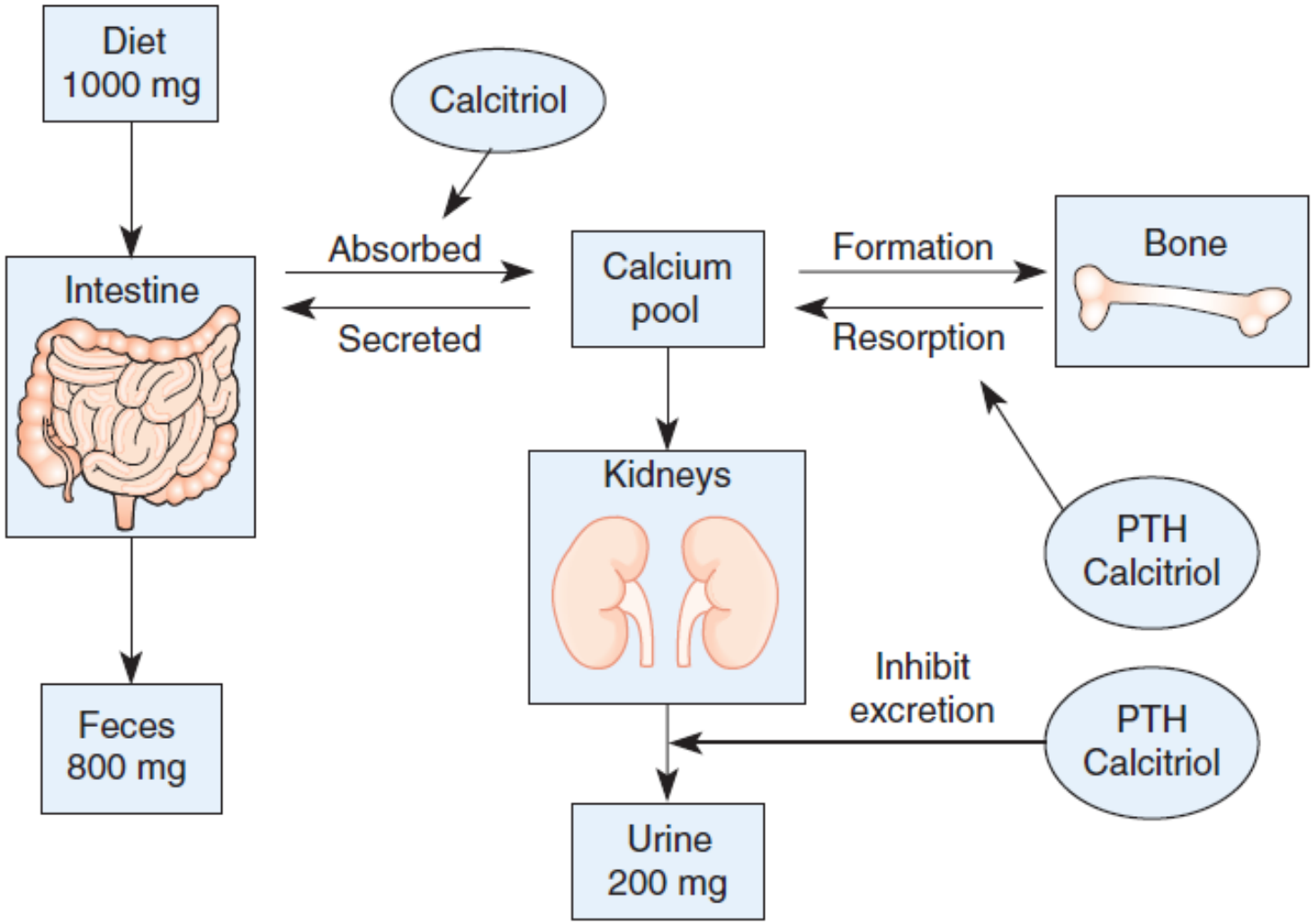




- PTH and FGF-23 decrease Pi reabsorption by decreasing the abundance of the sodium phosphate cotransporters.

(Blaine et al., 2014)

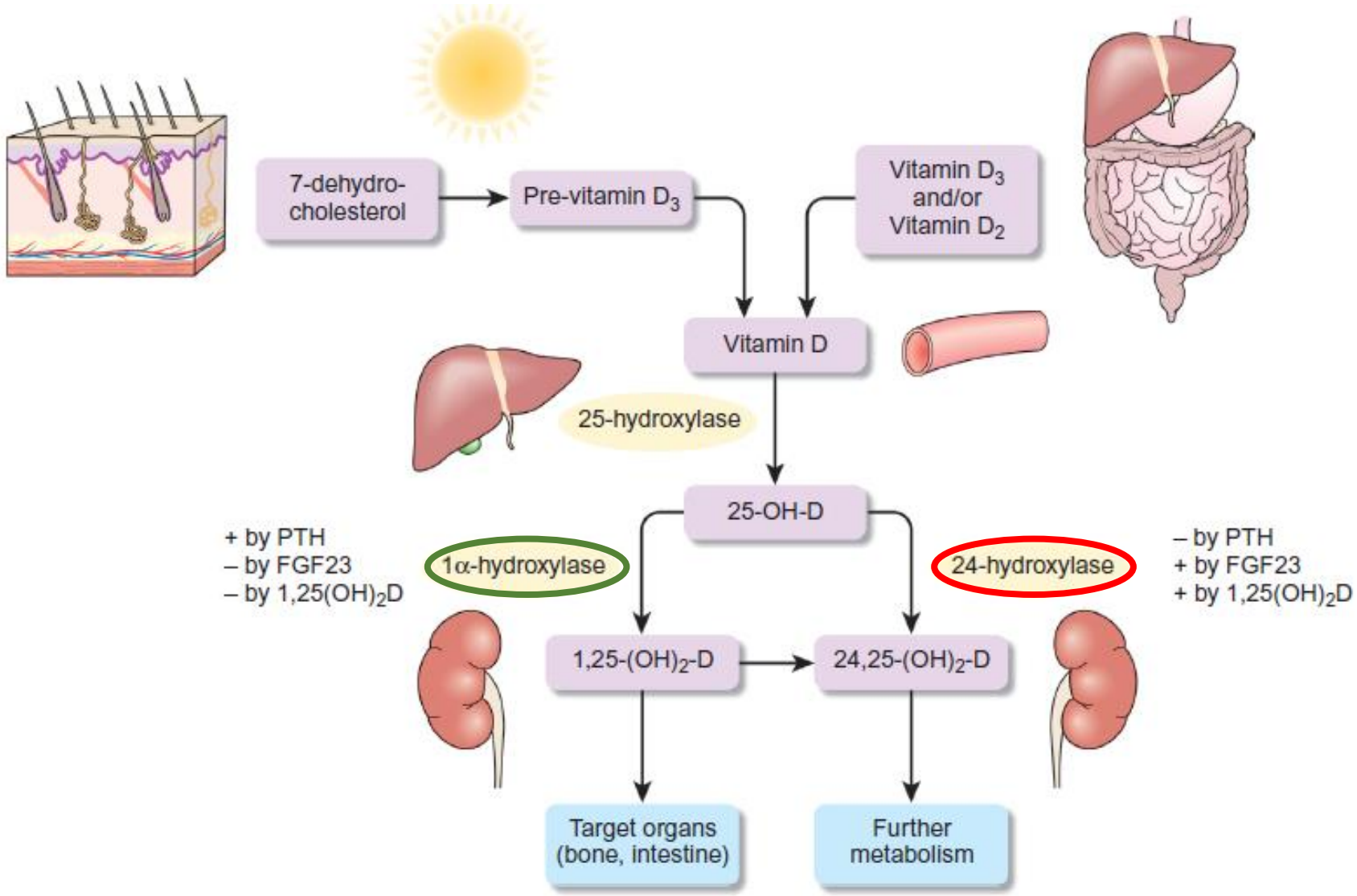
Hormonal control of Ca^{2+} and Pi homeostasis



• **Fig. 36.14** Overview of Ca^{++} homeostasis. PTH, parathyroid hormone. (From Koeppen BM, Stanton BA. *Renal Physiology.* 5th ed. Philadelphia: Elsevier; 2013.)

(Koeppen and Stanton, 2018)

Vitamin D metabolism



In the proximal tubules,

$25(OH)D_3$ is hydroxylated by 1α(OH)ase

>> $1,25(OH)_2D_3$ (calcitriol)

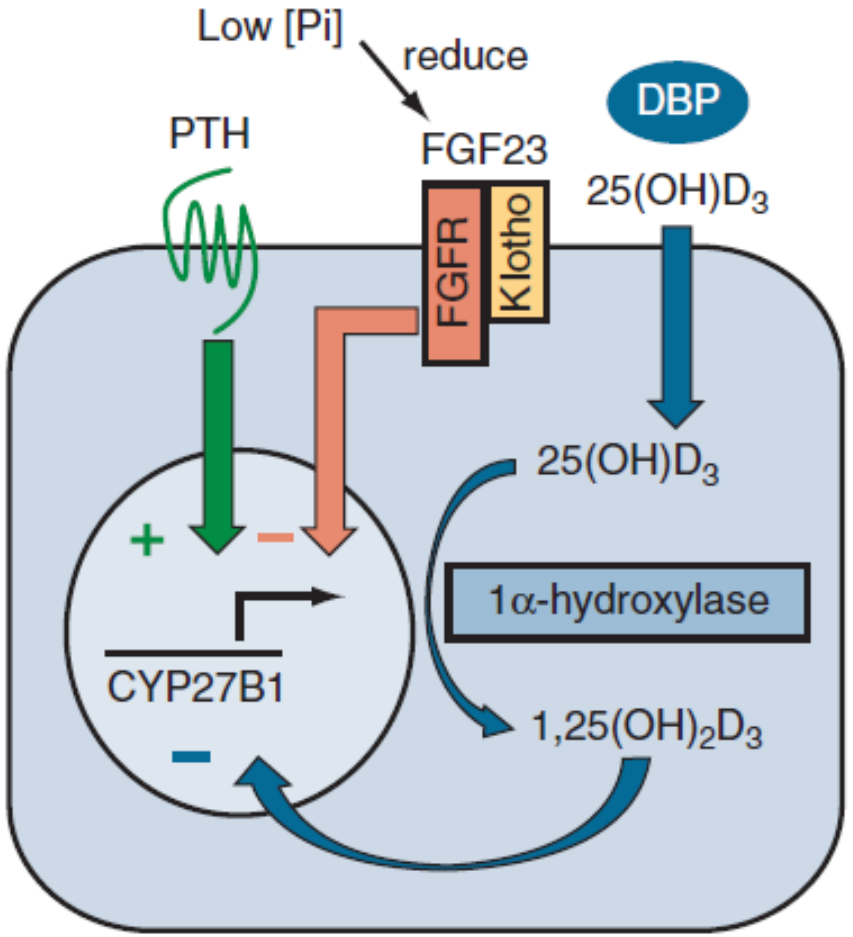
24(OH)ase catabolizes both $25(OH)D_3$ and $1,25(OH)_2D_3$

>> biologically inactive form

$24,25(OH)_2D_3$/$1,24,25(OH)_3D_3$

(DiMeglio and Imel, 2014)

Regulation of Vitamin D Production

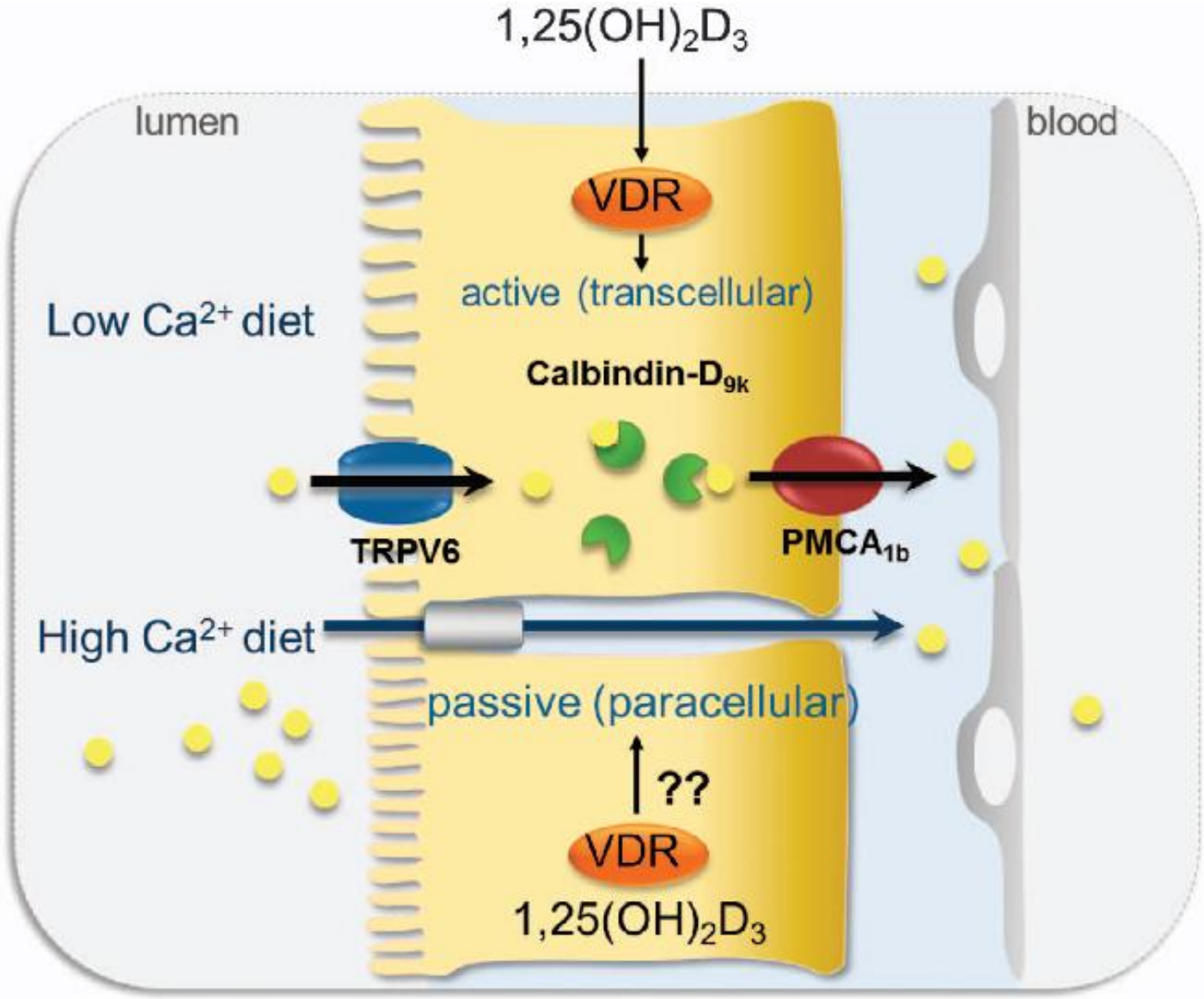


• **Fig. 40.7** Regulation of 1α-hydroxylase gene *(CYP27B1)* expression in the proximal tubule, showing stimulation by PTH and inhibition by FGF23 and 1,25-dihydroxyvitamin D. Hypophosphatemia probably stimulates 1α-hydroxylase by reducing FGF23 levels at least in part.

(Koeppen and Stanton, 2018)

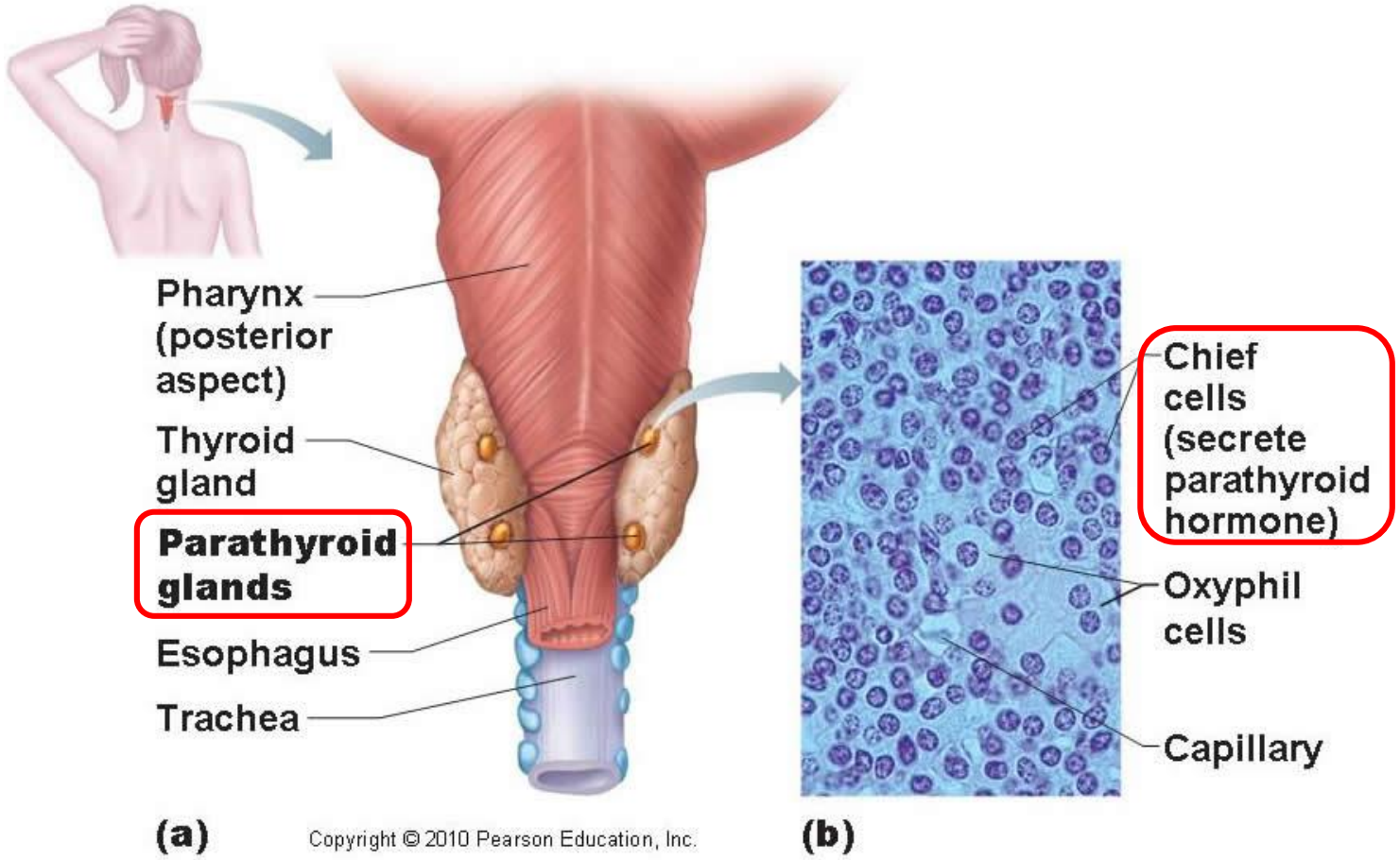
- **PTH stimulate $1,25(OH)_2D_3$ synthesis by activating 1α(OH)ase.**
- **Plasma Ca^{2+} and P_i levels can suppress $1,25(OH)_2D_3$ production in a negative feedback loop.**
- **FGF-23 inhibits 1α(OH)ase, but increases the expression of 24(OH)ase.**

Vitamin D regulates intestinal Ca^{2+} absorption



Parathyroid hormone (PTH)

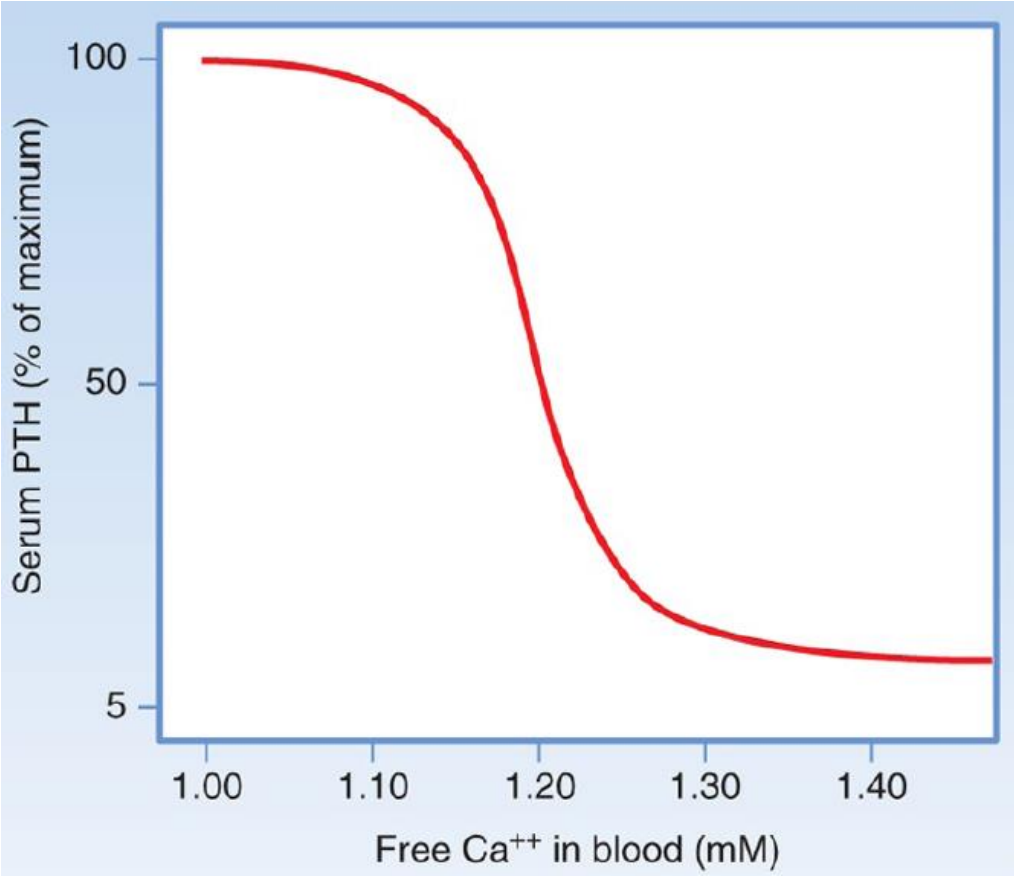
- a linear polypeptide containing 84 amino acid residues.
- secreted by **principal or chief cells** in the 4 parathyroid glands.



Regulation of PTH release

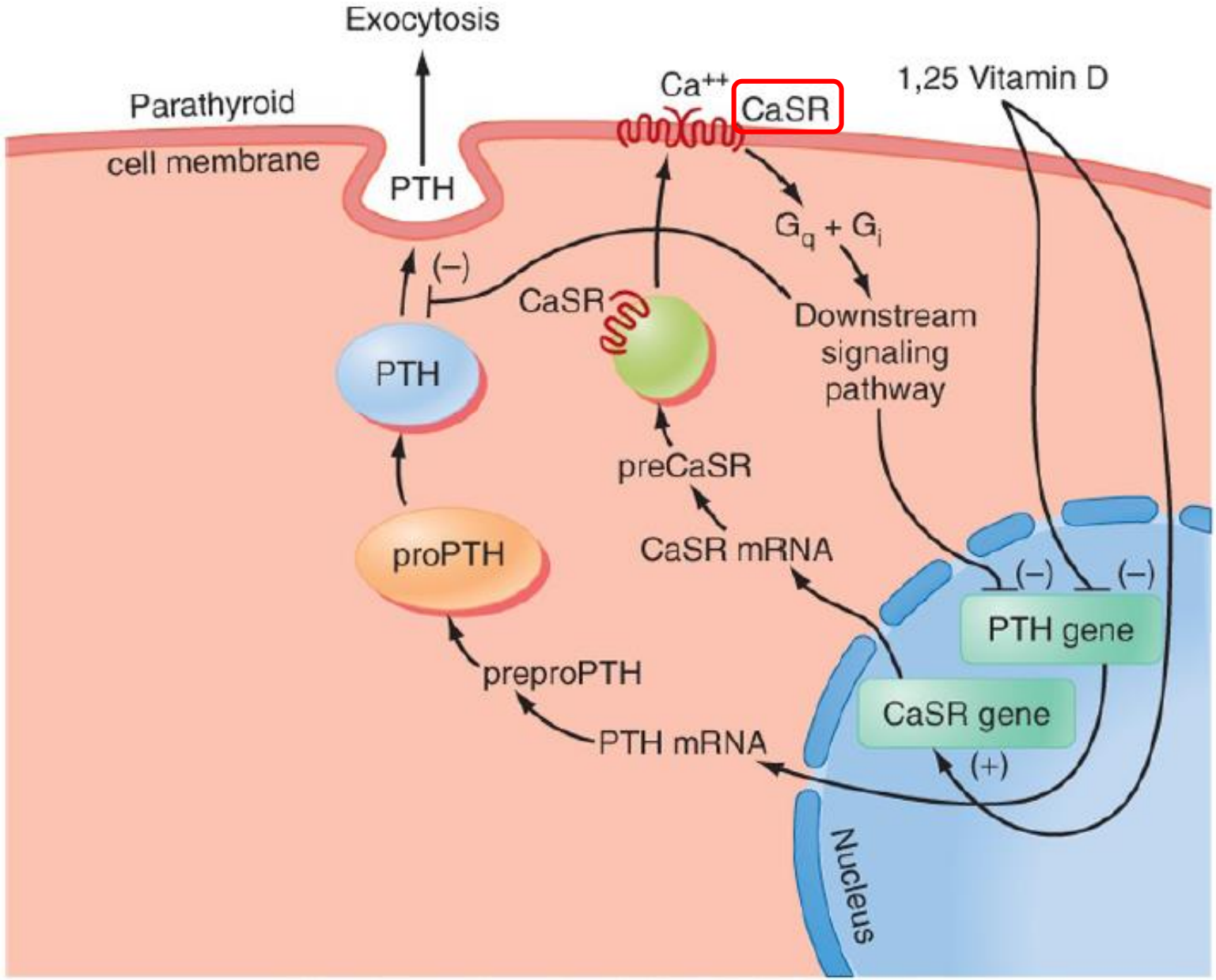
- Plasma ionized Ca^{2+} acts directly on these glands in a feedback manner to regulate the secretion of PTH.

- In hypercalcemia, PTH secretion is inhibited, whereas its secretion is stimulated by hypocalcemia.



(Koeppen and Stanton, 2018)

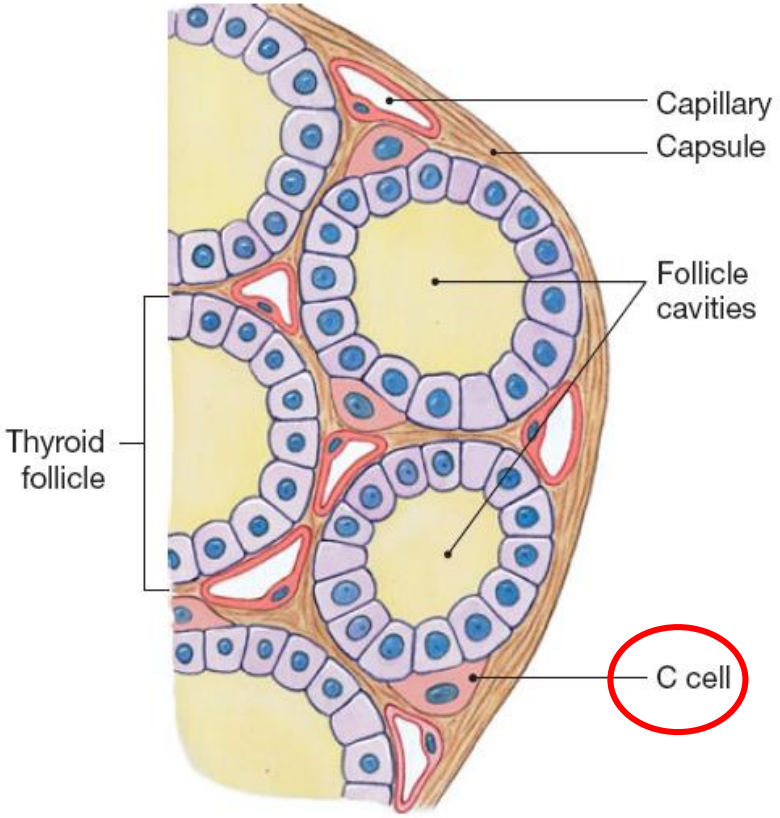
Regulation of PTH gene expression and secretion



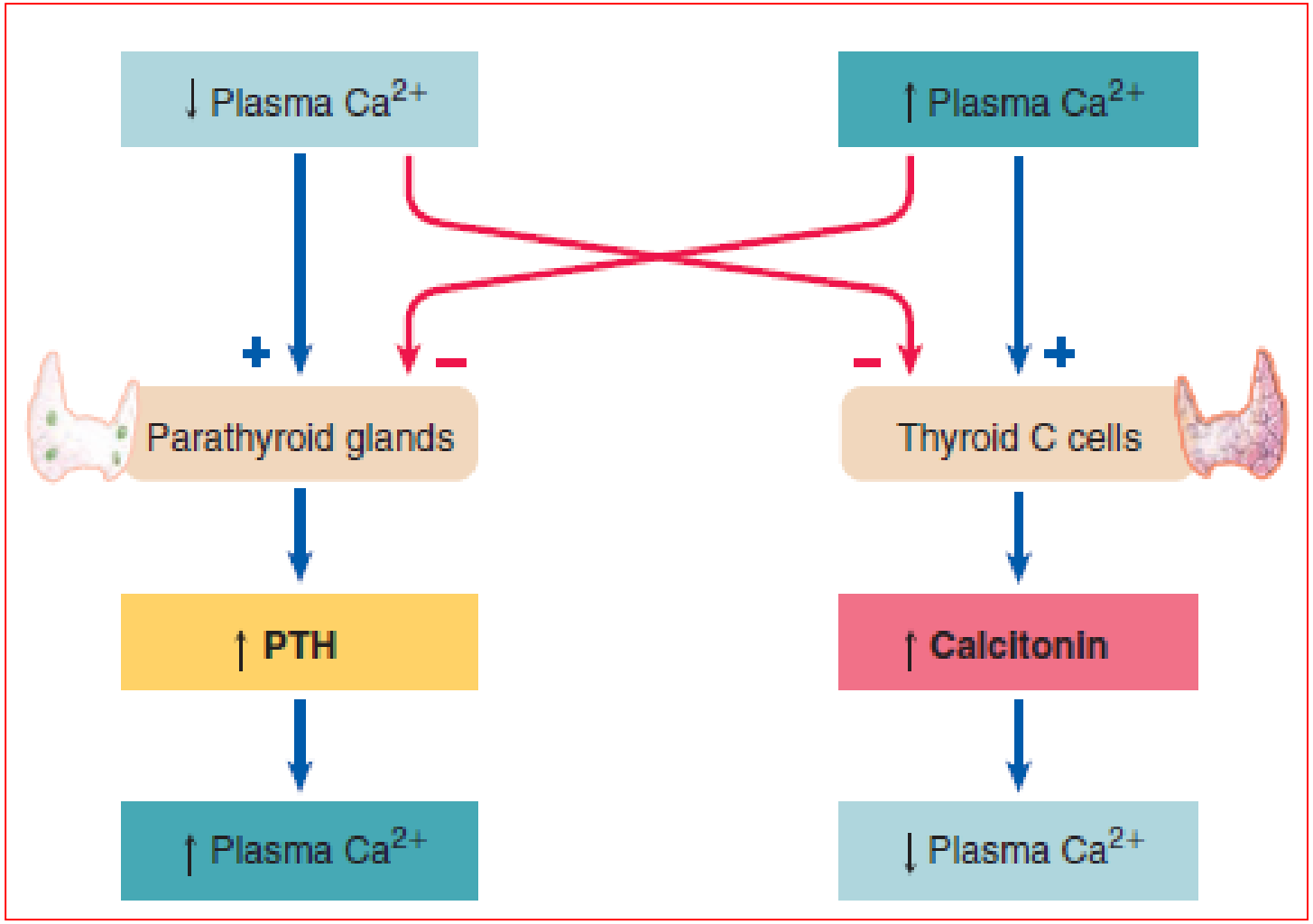
(Koeppen and Stanton, 2018)

Calcitonin

Calcitonin is released by parafollicular cells of the thyroid gland in response to increased plasma calcium levels



Parafollicular C cells



(Sherwood, 2012)

Fibroblast growth factor-23 (FGF-23)

- **Hypophosphatemic hormone >> Hypocalcemic hormone !**
- **Secretory protein from osteocytes and osteoblasts**

FGF-1 subfamily	FGF-1, -2
FGF-4 subfamily	FGF-4, -5, -6
FGF-7 subfamily	FGF-3, -7, -10, -22
FGF-8 subfamily	FGF-8, -17, -18
FGF-9 subfamily	FGF-9, -16, -20
FGF-11 subfamily	FGF-11, -12, -13, -14
FGF-19 subfamily	FGF-19, -21, -23

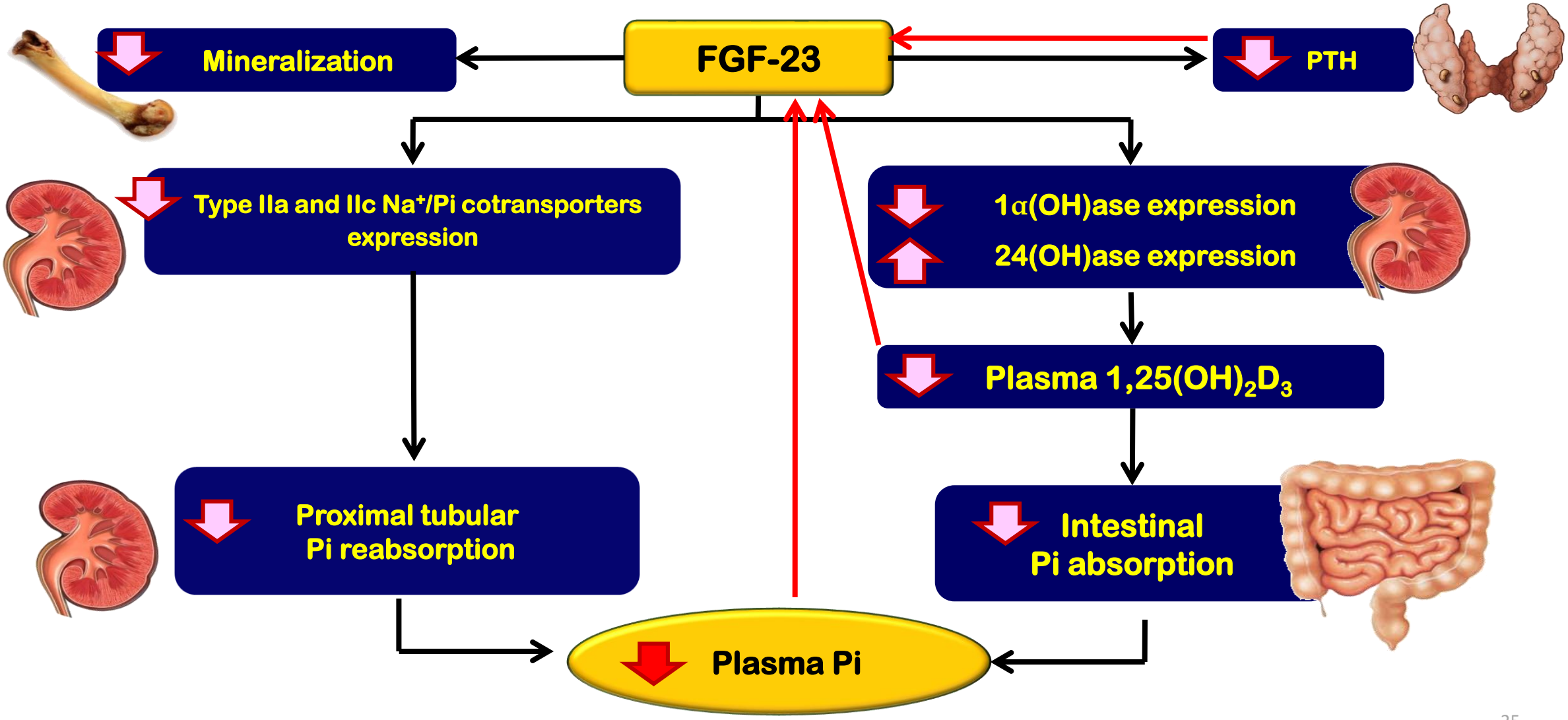
- Most FGFs — local factors
- FGF-23 — endocrine factor

- FGF23 directly controls the intestinal calcium absorption through both systemic and local (paracrine) mechanisms.

(Khuituan et al., 2012)

Functions of FGF-23

1α(OH)ase = $1,25(OH)_2D_3$ – producing enzyme
24(OH)ase = $1,25(OH)_2D_3$ – inactivating enzyme



Comparative actions of vitamin D, PTH and calcitonin

	Vitamin D	PTH	Calcitonin
Bone Osteoclasts: bone resorption Osteoblasts: bone formation	↑ Osteoclast activity (↑ bone resorption)	If constant ↓ Osteoblast activity (↑ bone resorption) If intermittent ↑ Osteoblast activity (↑ bone formation)	↓ Osteoclast activity (↓ bone resorption)
Kidney	↑ Calcium reabsorption ↑ Phosphate reabsorption	↑ Calcium reabsorption ↓ Phosphate reabsorption ↑ 1α(OH)ase synthesis	↓ Calcium reabsorption ↓ Phosphate reabsorption
Intestine	↑ Calcium absorption ↑ Phosphate absorption	(Indirect action) ↑ Calcium absorption ↑ Phosphate absorption	
Blood	↑ Calcium ↑ Phosphate	↑ Calcium ↓ Phosphate	↓ Calcium ↓ Phosphate

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Assigned e-learning !

Assigned e-learning

- Hypocalcemia
https://www.youtube.com/watch?v=t4dfHSTLInM (4 min)

- Hypercalcemia
https://www.youtube.com/watch?v=nko-JXU-c28 (4 min)

- คำถามท้ายบท